

5 Complex Systems Quiz ANS

1. What does the reductionist paradigm claim?

- A) Whole systems always exhibit properties that no analysis of their parts can ever explain
- B) Only statistical averages over many systems are meaningful in science
- C) Systems can be fully understood and described by understanding their parts
- D) Systems should never be studied by analyzing their individual components

ANS: C) Systems can be fully understood and described by understanding their parts

2. According to the deck, why can reductionist approaches not be used to study all real-world systems such as ecosystems and human society?

- A) Their components are so identical that studying them tells us nothing
- B) They contain no feedback loops that could be analyzed in isolation
- C) They are too small to be measured with modern instruments
- D) They have too many components with complex interdependencies

ANS: D) They have too many components with complex interdependencies

3. Which statement best defines emergence?

- A) A property that each individual component also has on its own
- B) A feature that appears only when a system is made smaller
- C) System-wide features that cannot be found in any single component
- D) A design choice imposed by a central controller on the parts

ANS: C) System-wide features that cannot be found in any single component

4. Which phrase captures why emergent features often "turn on" as a system scales up?

- A) Smaller is stronger: systems become safer as they shrink in size
- B) More is different: a quantitative change produces a qualitative one
- C) More is the same: adding components never changes behavior
- D) Faster is safer: higher speed removes emergent behavior altogether

ANS: B) More is different: a quantitative change produces a qualitative one

5. What does the deck's virology and wet-lab example illustrate?

- A) Combining two models always reduces the overall risk of misuse by spreading it out
- B) Two individually non-hazardous models can combine into an ultrahazardous one
- C) A model with only theoretical virology knowledge is already ultrahazardous on its own
- D) Hazard is determined entirely by the single most capable component in the system

ANS: B) Two individually non-hazardous models can combine into an ultrahazardous one

6. What is non-linearity?

- A) A change in input does not necessarily produce a proportional change in output
- B) Every output is exactly proportional to the input that the system receives
- C) A system that has no inputs coming in from its surrounding environment at all
- D) A feedback loop that always dampens any change that it encounters

ANS: A) A change in input does not necessarily produce a proportional change in output

7. Which pairing correctly matches the type of feedback with the deck's example?

- A) Positive feedback: the rich getting richer; negative feedback: homeostasis
- B) Positive feedback: a heater with a simple on/off switch; negative feedback: sand piles collapsing
- C) Positive feedback: shivering when cold; negative feedback: growing a colony
- D) Positive feedback: homeostasis; negative feedback: the rich getting richer

ANS: A) Positive feedback: the rich getting richer; negative feedback: homeostasis

8. What is self-organization?

- A) A central authority assigns each component a specific role in advance of operation
- B) Components direct themselves, producing collective properties without explicit instructions
- C) A system rebuilds itself from a stored blueprint after each failure it suffers
- D) Components wait for explicit external commands before they take any action at all

ANS: B) Components direct themselves, producing collective properties without explicit instructions

9. How does the deck define criticality?

- A) A system whose components are all identical to one another
- B) A system that has already been fully optimized for its task
- C) A system balanced at a tipping point between two different states
- D) A system that is completely stable and unable to change state

ANS: C) A system balanced at a tipping point between two different states

10. Which example does the deck give of the AI research community exhibiting self-organized criticality?

- A) Training only very small models on consumer laptops owned by individual researchers
- B) Releasing every model with open weights to the public as soon as it is trained
- C) Relying on NVIDIA GPUs, so a conflict in Taiwan could pause scaling
- D) Spreading training evenly across many hardware vendors in different countries

ANS: C) Relying on NVIDIA GPUs, so a conflict in Taiwan could pause scaling

11. What does distributed functionality mean for neural networks?

- A) Nodes or weights do not map to single concepts, so the network is harder to understand but possibly more robust
- B) The network's functions are neatly divided up between separate subsystems that can be studied alone
- C) Each neuron stores exactly one human-readable concept, so the network is easy to inspect
- D) The network cannot be trained on large datasets without losing most of its accuracy

ANS: A) Nodes or weights do not map to single concepts, so the network is harder to understand but possibly more robust

12. Which is the deck's example of power-law scaling in complex systems?

- A) A mechanical watch keeping constant time
- B) Kleiber's Law relating metabolic rate to body mass
- C) The proportional growth of a simple spring under load
- D) The ordering of cards in a sorted deck

ANS: B) Kleiber's Law relating metabolic rate to body mass

13. What is "risk compensation" in the discussion of adaptive behavior?

- A) Developers refuse to take any risk at all once a system becomes more reliable
- B) Insurance companies automatically pay for any AI accident that occurs in deployment
- C) Systems always become safer as soon as their reliability is improved by engineers
- D) Developers self-organize to their risk tolerance, rendering reliability advances ineffective

ANS: D) Developers self-organize to their risk tolerance, rendering reliability advances ineffective

14. Why does Lesson 1 say that armchair analysis is limited?

- A) Experiments are unnecessary once a sufficiently good theory has been developed
- B) Theory alone cannot anticipate all emergent properties, so some trial and error is required
- C) Failure modes can always be predicted completely by careful reasoning alone
- D) Theory is never useful for the purposes of AI safety research of any kind

ANS: B) Theory alone cannot anticipate all emergent properties, so some trial and error is required

15. In Lesson 2 (subgoals can supersede original goals), what does modularization refer to?

- A) A goal is broken into subgoals that may not be pursued in ways that fulfill the original goal
- B) A goal is handed to another agent whose own goals may clash with the original one
- C) A system is rebuilt from scratch every time that it is scaled up in size
- D) Several agents vote on a single shared goal before any of them acts on it

ANS: A) A goal is broken into subgoals that may not be pursued in ways that fulfill the original goal

16. What does Lesson 3 say about scaling up a safe system?

- A) A safe system always remains safe when it is scaled up to a much larger size
- B) Scaling up only improves a system's existing abilities and changes nothing else
- C) Scaling up removes emergent capabilities that were present at smaller scales before
- D) A safe system is not necessarily safe when scaled up, because new behaviors can appear

ANS: D) A safe system is not necessarily safe when scaled up, because new behaviors can appear

17. What does Lesson 4 conclude about building large, safe AI systems?

- A) A large AI system that is built from scratch is guaranteed to be safe in operation
- B) Gradual scaling of small systems always guarantees that the larger system is safe
- C) Complex systems are usually built directly from scratch by their original designers
- D) Working complex systems evolved from simpler ones, so scale up gradually from small, safe systems

ANS: D) Working complex systems evolved from simpler ones, so scale up gradually from small, safe systems

18. Which of the following does the deck give as an example of a wicked problem?

- A) Climate change
- B) Answering a simple math question
- C) Assembling furniture from instructions
- D) Solving a sudoku puzzle

ANS: A) Climate change

19. Which statement describes wicked problems?

- A) Exactly one correct result can always be found from the information provided to the solver
- B) There is always a clear endpoint, even if several approaches to the problem exist
- C) No single flawless solution exists, and it can be unclear when one has been solved
- D) They are found only in merely complicated systems that have no social elements in them

ANS: C) No single flawless solution exists, and it can be unclear when one has been solved

20. In the "Kill rats; present tails" example of intervention in a complex system, what was the unintended result?

- A) Energy use fell sharply as rat tails were collected
- B) The rat population was permanently eliminated from the city
- C) Rats became easier for the authorities to control and count
- D) People bred rats on farms to collect the reward

ANS: D) People bred rats on farms to collect the reward